

Department of Computer Science and Engineering, RNSIT

AI & ML Track – 4-Week Internship Schedule, 2026

Week	Day	Topic	Morning Session (Theory)	Afternoon Lab (Practical)	Evening Task / Assignment
1	1	Orientation & Python/ML Setup	AI vs ML vs DL hierarchy; Python ML ecosystem; Colab setup	Setup Google Colab; run first ML notebook; explore Keras datasets	Watch Andrew Ng's ML Specialization Week 1 videos; take notes
	2	Mathematics for ML	Linear algebra recap (matrices, eigenvectors), calculus (gradient, chain rule), probability	Lab: NumPy matrix operations; implement gradient descent from scratch	Solve 10 linear algebra problems; visualise gradient descent in Python
	3	Classical ML – Regression & Classification	Linear/Logistic Regression; cost function; gradient descent; Scikit-learn pipeline	Lab: Predict house prices (Linear Regression) + classify emails (Logistic Regression)	Tune regularization (L1/L2) on both models; document effect on accuracy
	4	Trees & Ensemble Methods	Decision Trees, Random Forest, XGBoost, Gradient Boosting; feature importance	Lab: XGBoost classifier on Titanic; SHAP values for explainability	Kaggle: submit predictions for Titanic competition; screenshot leaderboard rank
	5	Model Evaluation	Cross-validation, confusion matrix, ROC-AUC, precision-recall curves, overfitting	Lab: Comprehensive evaluation of Week models; plot learning curves	Week 1 quiz; identify project domain (CV, NLP, tabular, or RL)

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2	6	Neural Networks Fundamentals	Perceptron, MLP, activation functions, forward/backward propagation, loss functions	Lab: Build a 3-layer NN from scratch in NumPy; classify XOR problem	Implement sigmoid, ReLU, tanh from scratch; plot activation curves
	7	Deep Learning with Keras/TensorFlow	Keras Sequential & Functional API; layers, optimisers, callbacks, ModelCheckpoint	Lab: Build image classifier on MNIST using Keras CNN; achieve >98% accuracy	Industry expert session: 'AI in Healthcare/Finance' – Q&A writeup
	8	Convolutional Neural Networks (CNNs)	Conv layers, pooling, stride, padding, VGG/ResNet architecture; transfer learning	Lab: Fine-tune VGG16/ResNet50 on a custom 5-class image dataset (Colab T4)	Kaggle Dogs vs Cats: transfer learning submission; document results
	9	Recurrent Neural Networks & NLP	RNN, LSTM, GRU; tokenisation, word embeddings (Word2Vec, GloVe); sentiment analysis	Lab: Sentiment analysis on IMDb using LSTM; compare with Logistic Regression baseline	Complete Coursera NLP in TensorFlow Week 1; submit screenshot
	10	Project Proposal	Review proposals: image classification / NLP / anomaly detection / generative	Lab: Dataset prep for project; baseline model run	Submit proposal: problem, dataset, baseline accuracy, target accuracy, plan

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3	11	Transformers & Hugging Face		Attention mechanism, BERT, GPT architecture; Hugging Face pipeline & fine-tuning	Lab: Use HuggingFace pipeline for text classification, NER, summarisation on project data	Fine-tune a DistilBERT model on your domain dataset (Colab)
	12	Computer Vision Advanced		Object detection (YOLO intro), image segmentation, OpenCV – face detection, video processing	Lab: Run YOLOv8 on a custom image set; implement face detection pipeline	Write report Section 3 (Architecture & Implementation)
	13	Generative AI & LLMs (intro)		GANs, VAEs, diffusion models overview; prompt engineering; LLM APIs (Gemini/OpenAI)	Lab: Build a simple chatbot or text-to-image prompt pipeline for project	Industry expert review: present project; get feedback on model design
	14	MLOps & Deployment		Model saving (ONNX, TF SavedModel), FastAPI serving, Gradio demo, Hugging Face Spaces	Lab: Deploy project model as a Gradio app on Hugging Face Spaces	Write report Sections 4–5; ensure Gradio app is public
	15	Internal Demo		Present model demo to faculty (10 min per group); metrics & architecture explanation	Lab: Incorporate feedback; polish app	Prepare final demo slides; write report conclusion

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4	16	Explainability & Ethics		SHAP, LIME, responsible AI, bias in datasets, fairness metrics	Lab: Apply SHAP to your project model; identify top 5 features	Write a 300-word reflection on ethical implications of your AI model
	17	Mini-Paper Writing		IEEE format: abstract, related work, methodology, experiments, results, conclusion	Lab: Write mini-paper draft; include accuracy tables and training plots	Peer-review partner's paper; finalise your own; submit both
	18	Final Demo Day		Live demo of AI model (Gradio/Streamlit) to panel (10 min + Q&A)	Panel evaluation	Submit: GitHub, Hugging Face Space link, report (PDF), mini-paper
	19	Viva Voce		Individual viva: architecture choices, hyperparameters, why this model? (7 min)	–	Prepare: 'Explain back-propagation', 'Why did you choose CNN over RNN?'
	20	Valediction		Awards, certificates, talk: 'AI Research & Industry Roles'	Portfolio, next steps: Kaggle, research papers, internships	Publish Kaggle notebook; write a LinkedIn post about your AI project